

Statistical Visualization and Outlier Detection with Pandas

Task 1 – Sales Comparison Analysis

- a. Load monthly sales data for two stores.
- b. Perform descriptive statistical analysis for both datasets.
- c. Generate appropriate plots for visual comparison.
- d. Interpret differences between the two stores based on numeric and visual findings.

Task 2 – Outlier Detection in Housing Data

- a. Plot a histogram to visually inspect the presence of outliers in the price_per_sqft column.
- b. Determine the value ranges that include 95% and 99.7% of data points.
- c. Apply the Interquartile Range (IQR) method to remove outliers and display the resulting DataFrame.
- d. Report the size of the DataFrame after IQR-based filtering.
- e. Use the 3 standard deviations method to remove outliers and display the result.
- f. Plot histograms and overlay bell curves for the data from steps c and e.
- g. Reapply outlier detection using a z-score threshold of 3 and confirm the results.

Task 3 – Wine Quality Dataset Analysis

- a. Check and remove any missing or duplicate records.
- b. Compute and visualize the correlation matrix using a heatmap.
- c. Count the frequency of each wine quality level and display it as a bar chart.
- d. Plot histograms for all variables and identify potential outliers.
- e. Create a box plot (catplot) comparing wine quality and alcohol content.
- f. Generate a scatter plot of Alcohol vs. pH, color-coded by wine quality.

Task 4 – Iris Dataset Analysis

- a. Compute mean, median, min, max, and standard deviation of petal and sepal ratios per species.
- b. Generate box plots to show petal and sepal ratio distributions per species.
- c. Analyze the correlation between petal and sepal features based on scatter plot and correlation coefficient.
- d. Plot histogram for sepal.length and interpret its distribution.
- e. Determine the range of values covering 68% and 95% of the sepal.length data.
- f. Calculate and plot mean and standard deviation of sepal.length and sepal.width for each species as bar graphs.